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An evaluation of multi-effect desalination with a thermal vapour compression system in terms of thermo-economics

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Abstract: In this study, a multi-effect distillation system (MED-TVC) that is presently in use in a thermal power plant was analysed. The operating data of an existing MED-TVC (2,000 m³/day) desalination process were used for the present analysis. The analysis of the considered system was performed based on the energy, exergy and thermoeconomic approach. Also, the effect of operating plant load, seawater temperatures and upper brine temperatures on the performance of the desalination plant was investigated. The gain output ratio and exergy efficiency increased with increasing production load and seawater temperature. When the production load and seawater temperatures were 100% and 33°C, respectively, the gain output ratio and exergy efficiency obtained was 7.73 and 2% respectively. The results showed that the steam ejector was responsible for about 45% of the total exergy destruction. The unit product cost of the distilled water at the full load was calculated by 1.7 \$/m³.

Keywords: brine temperature; desalination; energy; exergy destruction; exergy efficiency; gain output ratio; multi-effect; sea water temperature; thermo economic analysis; thermal power plant; thermal vapour compression.

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1 Introduction

In recent years, industrial growth, economic development, and improvement in living standards in parallel with the growth of the world's population have led to a substantial increase in global energy demand. Besides, the increasing consumption of fossil fuels for energy production to meet energy demands has led to increased climate change. This trend has also created infrastructure creating widespread and long-term problems for the planet (Mohammadi and McGowan, 2018). In this context, increasing the energy efficiency of thermal energy systems, as well as achieving energy savings is the essential global challenge (Ghaebi and Abbaspour, 2018). Today, most of these facilities were built at the seaside to meet the water demand for thermal power plants more economically. It should be noted that in these plants, energy efficiency is significantly dependent on seawater temperature, and the amount of water used in the condenser unit is enormous. Today, since clean water production has become increasingly important in many countries, the multi-effect desalination (MED) method has become an alternative parallel method for clean water production as well as generating electricity by integrating MED units into steam-powered plants (Askari et al., 2018).

Depending on the world population growth, an increase in clean water demand combined with occasional droughts around the planet, necessitate the discovery of new and alternative clean water resources (Khalid et al., 2018). Desalination has become one of the world's most important, traditional water processes in recent years and is especially important in areas with less water (Zarzo and Prats, 2018). Desalination is one of the separation methods involving a large number of components of various sizes using a large amount of energy. The losses in these units are the main issues for both the design and operation of the plants. Separating salts from salty water is a costly process because it consumes much more energy compared to other freshwater supply and treatment options (Li et al., 2018). In order to be economically viable in comparison to other clean water procurement methods, this method needs to bring down the energy demand for water treatment processes such as removing salts from the seawater (Fitzsimons et al., 2015; Brogioli et al., 2018).

Therefore, a large number of scientific and technological research studies have been applied to enhance the performance of distillation plants, in particular, to improve the performance of seawater distillation (or removal of salt) saltwater technologies. Al-Mutaz and Wazeer (2014) investigated three different configurations of the multi-effect evaporator system from a different perspective than those works reviewed above which are parallel/diagonal, forward and reverse configurations. They obtained results that revealed that the difference in vapour temperature and temperature per effect for three

different feed methods had a significant effect on the specific heat transfer area of the effects. It was concluded that the parallel/cross-feed method had better performance characteristics than the other two feed methods. It is possible to have a higher gain output ratio if the parallel/cross-water feed method is used instead of a forward and backward water feed method in order to achieve less specific heat consumption.

Beyralvand and Yaghoubi (2018) performed thermoeconomic analysis and multi-objective optimisation of MED-TVC and MED-TVCFH systems using artificial bee colony algorithm. Systems were optimised for the highest exergy efficiency and lowest cost of the produced desalted water. The results illustrated that by choosing final optimum solutions, the desalted water cost declined by 18.1% and 28.8% for MED-TVC and MED-TVCFH systems respectively. Also, exergy efficiency improved from 3.2% and 4.04% in the main case to 3.63% and 4.47% for MED-TVC and MED-TVCFH respectively in the optimal situation. Sekar and Saravanan (2011) carried out an exergetic analysis for vapour absorption heat transformers integrated with MED system. In their study, a simulation model has been developed to evaluate the exergetic efficiency, exergy losses in the components and exergy balance of the system.

Fellah et al. (2014) conducted the thermodynamic analysis of a unit of the Zawia city desalination plant. The plant composed of four units with a total capacity of 80,000 m³/day and MED-TVC technology. In the summer and winter analysis sessions, design and real-time data were accepted. High levels of exergetic destruction occurred during the thermal vapour compression processes, contributing to the mean value of 45.76% of the amount of exergy consumed. Losses corresponded to approximately 22.25% of the consumed exergy. Effects, pre-heaters and the end condenser provided a mean value contribution of 16.13%, 7.84% and 8.02%, respectively, for consumed exergy. Shen et al. (2011) studied a low-grade temperature multi-effect distillation plant. In addition, the influence of the motive steam parameters on the performance parameters of the MED-TVC desalination system was evaluated for variable steam pressures and temperatures. With higher steam pressure, the performance ratio of the desalination plant achieved higher values. Also, it has been reported that the recirculation position of vapour in a MED system substantially affects the GOR.

The efficiency of processes is generally a function of the energy efficiency; however, this does not indicate where energy inefficiencies and actual energy wastes are located or that revisions must be made to increase the overall use of energy in installations. However, it is necessary to use the second law analysis which is known as the exergy analysis in order to define exact energy losses (Mabrouk et al., 2007b; Sekar and Saravanan, 2011). This study presents energy and exergy analyses and exergoeconomic study of four-effect desalination processes at different operating conditions. The originality and contributions of this study are as follows:

- 1 showing key performance parameters of the overall system considering variable production loads, different seawater temperatures and upper brine temperatures
- 2 defining exergy destruction of each subsystem
- 3 displaying unit cost of distilled water by means of exergoeconomic analysis
- 4 determining most efficient operating points of the desalination plant.

In this study, technical data were used by taking from existing steam power plant. In this study, measurements such as pressure, temperature and flow rate were made by

instruments with high accuracy. These measurements are calibrated each month by maintenance teams. Moreover, they are also controlled by the internationally accredited company for certain periods.

2 Material and methods

2.1 Energy and exergy evaluations

The energy and exergy analyses on the MED-TVC were carried out by conducting mass and energy conservation laws to subsystems such as end condensers, evaporators and vapour ejectors. Some assumptions were made in applying the analysis: steady-state operation, equal temperature difference throughout the effects, distillation (without distillation contributions from each effect), and specific temperature changes as well as temperature, salinity and boiling point elevation were ignored (Bin Amer, 2009).

Most of the thermodynamic analyses carried out on steam plants and TVC systems are based on the first law of thermodynamics. Since both thermal, potential and kinetic energy are used in steam power plants, the first law of thermodynamics has a special significance. Although the first law of thermodynamics is an essential method of evaluating the overall performance of the thermal vapour compression desalination plant, this analysis does not add to the quality of the transferred and converted energy. A thermodynamic analysis using the first law could not illustrate where the maximum loss of available energy happens and concludes that energy loss to the environment and blowdown are the only considerable losses. In this sense, exergy analysis is necessary to place all energy interactions on the same basis and to provide appropriate guidance for process improvement (Alasfour et al., 2005). The TVC system performance parameters are as follows. The gain output ratio is presented as:

$$GOR = \frac{D_t}{S} \quad (1)$$

where D_t and S represent the mass flow rates of the total distilled water and motive steam supply respectively. The specific heat consumption per unit of distillate is expressed by:

$$\Upsilon_{en} = \frac{S(h_s - h_c)}{D_t} \quad (2)$$

where h_s and h_c are specific enthalpies of the motive steam and the condensate in the environmental condition. The definition of specific exergy destruction is:

$$\Upsilon_{ex} = \frac{E_s}{D_t} = \frac{S[(h_s - h_c) - T_0(s_s - s_c)]}{D_t} \quad (3)$$

where E_s and s_s designate exergy and entropy of the motive steam. Here, s_c is the entropy of the outlet condensate as saturated liquid at the ambient temperature.

The total loss of exergy consists of the only exergy losses of the desalination system components. The detail of energy and exergy balances for the steam ejector, effects and end condenser can be found in the studies of Alasfour et al. (2005) and Alasfour and Alajmi (2010).

The total desalinated product output is expressed by the following:

$$D_t = \sum D_i = D_1 + D_2 + \dots + D_n \quad (4)$$

Exergy efficiency is employed as a measure of the thermodynamic reversibility of a desalination system. Exergy efficiency (η_{11}), measures the extent of irreversible losses within a system. Bejan et al. (1996) define exergy efficiency as the ratio of the useful exergy of the outputs of the system to the exergy of the system inputs:

$$\% \eta_{11} = \frac{E_D}{E_{in}} \quad (5)$$

2.2 Thermo-economic analysis

Thermo-economics is an engineering discipline that combines thermodynamic assessments on an exergy basis in a manner that is economically appropriate. At the level of system components, it provides information that is useful for designing and operating a cost-effective system, which is obtained through traditional exergy, exergy and economic analyses.

For each flow in a plant, a parameter called flow cost rate $\dot{C}$ (\$/h) was determined, and a cost rate balance is expressed for each elements as follows:

$$\dot{C}_s + \dot{C}_f + \dot{Z} = \dot{C}_l + \dot{C}_d \quad (6)$$

$\dot{C}_s$, $\dot{C}_f$, $\dot{C}_l$, $\dot{Z}$ and $\dot{C}_d$ are flow cost rate of motive steam, feedwater, lost, desalination plant and distillate respectively. Figure 1 helps the location of cost rate of each stream. Exergy costing, a cost is associated with each exergy stream. Cost balance equation associated with exergy of each stream is shown as follows:

$$c_s E_s + c_f E_f + \dot{Z} = c_l E_l + c_d E_d \quad (7)$$

where c_s , c_f , c_l and c_d are the cost per unit of exergy of related stream in \$/kJ. The unit product cost is shown by equation (8) (Mabrouk et al., 2007a):

$$Unit\ cost \left(\frac{\$}{m^3} \right) = \frac{\dot{C}_l + \dot{C}_d}{D_t} \quad (8)$$

Fuel and product exergy must be identified in this assessment. When the product exergy is defined based on the component in question, the fuel refers to the consumed source in the production. The product and fuel are shown in terms of exergy. The cost ratios associated with the fuel and product of a component are obtained by changing the rates of exergy.

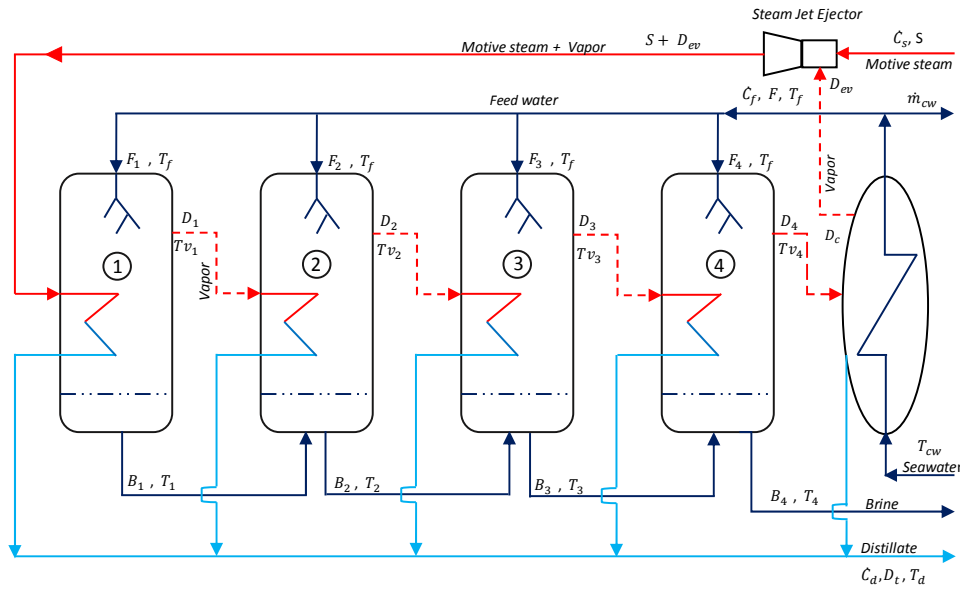
$$\dot{Z} = \frac{Z.CRF.\phi}{N \times 3,600} \quad (9)$$

Here, the procurement cost of the component is shown as Z , and the capital recovery factor is shown as CRF . CRF is defined based on the interest rate and equipment lifetime, and is defined as shown below (Bejan et al., 1996):

$$CRF = \frac{i \times (1+i)^n}{(1+i)^n - 1} \quad (10)$$

The interest rate is shown as i and the total working period of the plant in years is shown as n . Additionally, N refers to the annual number of unit operating hours and ϕ is the factor of maintenance, usually given as 1.06.

Figure 1 A schematic diagram of the MED-TVC system (see online version for colours)



3 Results and discussion

3.1 Plant description

The existing desalination plant has been commissioned to fulfil the water requirements of the steam power plant. The plant is located in the eastern part of the Mediterranean region in Turkey. Sea water salinity is about 44,000 ppm at the region where the plant is built. It is based on MED with thermal vapour compression technology and comprises four effects with 2,000 m³/day capacity in total at full load. A schematic diagram of the MED-TVC system is presented in Figure 1. The plant comprises four horizontally falling film evaporators. The incoming seawater with 117.5 kg/s is preheated in a distiller cooler and an end condenser. A fraction of the feed seawater ($\dot{m}_{cw}$) is returned to the sea when necessary and the 83.3 kg/s is separated into four equal flows to effects.

The turbine removes a small proportion of the live steam at 15 bar and delivers into the vacuum ejector, and the remaining steam is transferred to the steam ejector as motive steam. The flow of steam from the nozzle forms a vacuum that draws the steam formed into the final effect. The combined vapour streams are then compressed in an ejector to

meet the first thermal requirement. The condensate leaving the first effect is divided into two streams. The heated steam causes the feed from the top to evaporate the brine. The steam generated by the first effect is then fed to the tube side of the next effect, which features the same condensation and a repetition of the evaporation process. The brine produced in the first effect is stepped down to the next effect for flashing. The upper brine temperature (T_1) is at about 66°C in the first effect and last effect brine temperature (T_4) at about 54°C. Temperature drop (ΔT) across effects are assumed to be equal to 4°C. In the last effect, seawater is discharged as brine. Distilled water collected from the four effects is fed to the water to the steam cycle of the steam power plant at about 42°C as a distillate temperature (T_d) (Hamed et al., 1996).

3.2 Results of the thermo-economic analysis

A thermo-economic analysis was applied to an existing MED-TVC system operating in Turkey. The variation in gain output ratio as a function of production load is presented in Figure 2(a). As seen from this figure, the gain output ratio increased with increasing production load and seawater temperature. When the production load and seawater temperatures were 100% and 33°C, respectively, the gain output ratio obtained was 7.73. When the seawater temperature rises, the steam required decreased to increase its temperature to the upper brine temperature, and as a result, the GOR value increases.

Variation of the exergy efficiency of the overall desalination plant as a function of production load is presented in Figure 2(b). Results revealed that the exergy efficiency of the overall desalination plant increases with increasing production load and seawater temperature. The desalination process is an energy-intensive process. Accordingly, high-quality energy is converted into low-quality energy, as motive steam energy is converted into distilled water so that the exergy efficiency of the system is around 1 to 2%, which increases with rising operating load and increasing seawater temperatures.

Variation of the specific heat consumption as a function of production load for different seawater temperatures is shown in Figure 2(c). Calculations showed that the specific heat consumption decreased with increased seawater temperatures. Also, the reduction in the production load across the system, which can be achieved by decreasing the rate of flow of motive steam, increases the specific heat consumption and, as a result, decreases the specific exergy consumption. The specific enthalpy of the motive vapour increases in parallel with the rising pressure. The increased rate of specific enthalpy is lower compared to the decreasing amount of steam required so that the specific heat consumption is reduced.

Specific exergy destruction is associated with an assessment of the performance. Variation of the specific exergy destruction as a function of production load for different seawater temperatures is shown in Figure 2(d). The specific exergy destruction decreased with increased production load and seawater temperature. The desalination process is an inefficient process. Too much energy is destroyed when the steam flows through the ejector jet. Also, the steam energy disappears during heat transfer into different effects.

Figure 3(a) presents the variation of the gain output ratio as a function of upper brine temperature for different seawater temperatures. As seen from this figure, the gain output ratio is inversely proportional to the upper brine temperature. As the upper brine temperature increased, the vacuum pressure of the effect was adversely affected.

Therefore a high amount of energy is needed to evaporate sea water. In the system, the potential for scale formation, corrosion, and pretreatment costs are reduced by a lower brine temperature.

Figure 2 Variations of, (a) the gain output ratio (b) the exergy efficiency of the overall plant (c) the specific heat consumption (d) the specific exergy destruction as a function of production load for different seawater temperatures (see online version for colours)

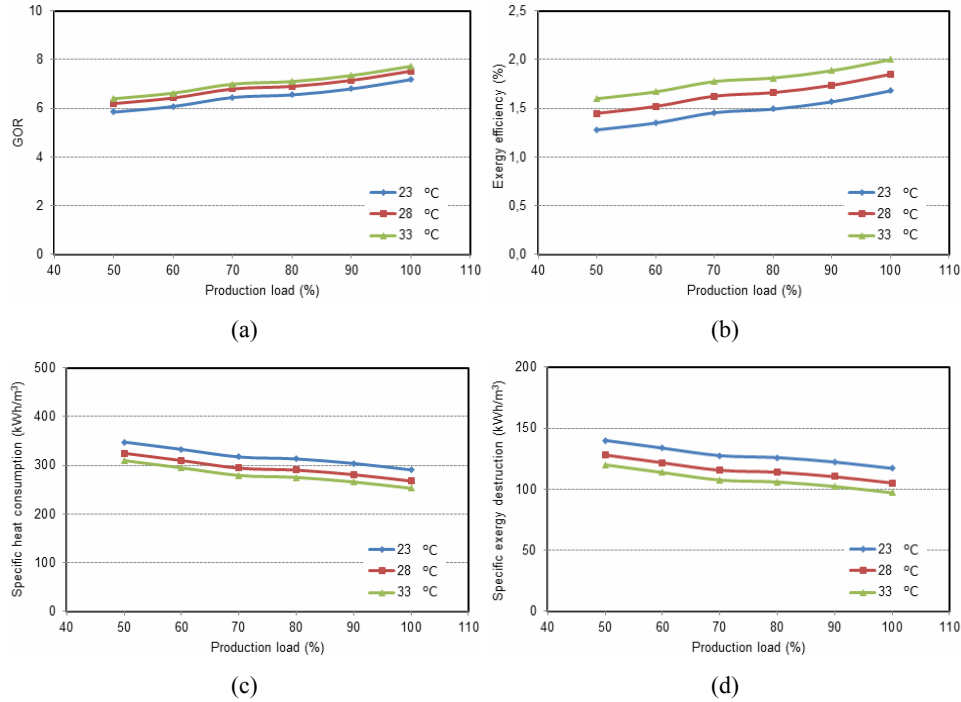


Figure 3(b) shows the variation of the specific heat consumption as a function of upper brine temperature. The obtained results have revealed that as the upper brine temperature increased, the specific heat consumption significantly increased for all three seawater temperatures. So, the specific heat consumption is directly proportional to the upper brine temperature. In the desalination process, the heat transfer area is decreased when the upper brine temperature is rising. Therefore, the heat transfer rate gets lower and a higher amount of energy is needed.

Variation of specific exergy destruction as a function of upper brine temperature is shown in Figure 3(c). The increase in the specific total exergy destruction is noteworthy as the upper brine temperature increases in the three configurations because of an increase in the ejector compression ratio, which leads to higher irreversibilities.

The subsystem that causes exergy destruction in the desalination system is the steam jet ejector, due to the high temperature and high pressure of its heat input. In the TVC system, the steam jet ejector is responsible for a loss of 45% of the total input exergy, with the second highest loss of exergy (20%) due to the effects. Subsystem exergy destruction ratios at full load and a temperature of 23 °C are shown in Figure 4.

Figure 3 Variations of, (a) the gain out ratio (b) the specific heat consumption (c) the specific exergy destruction as a function of upper brine temperature for different seawater temperatures (see online version for colours)

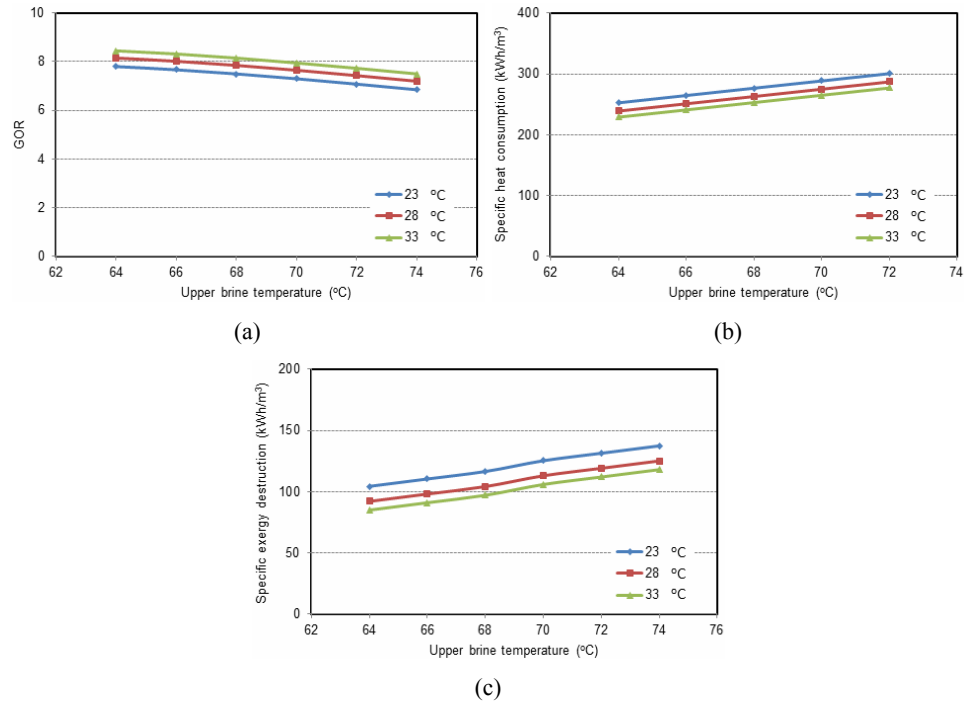
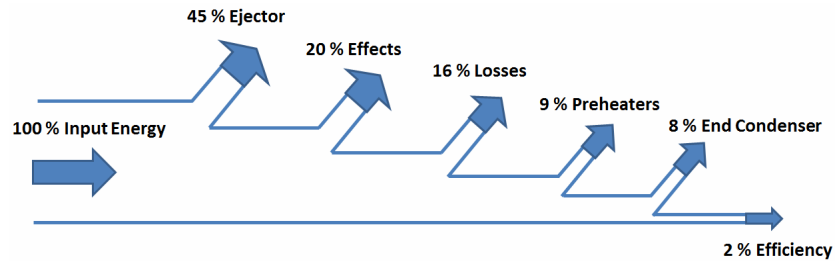


Figure 4 Exergy destruction ratios of the subsystem at full load (see online version for colours)

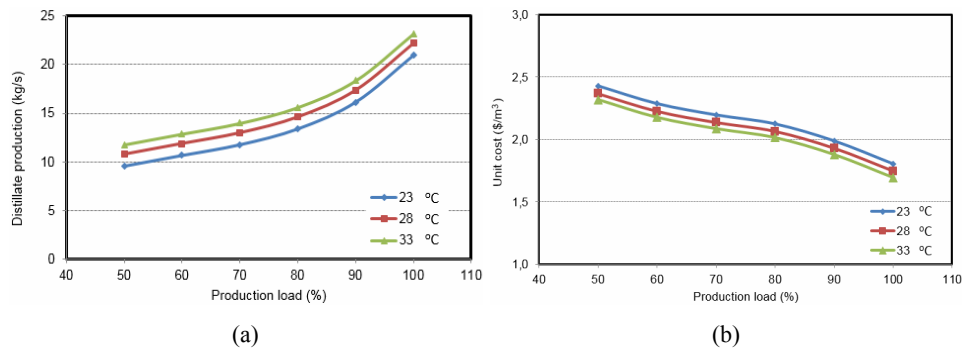


The control of the distilled water outlet can be maintained by regulating the flow rate of the motive steam to the thermal vapour compressor. The increase in the motive steam flow rate will automatically cause an increase in the distilled water output [Figure 5(a)]. Calculations indicated that as the production load and seawater temperature increased, the produced water significantly increases. The increase of production load from 50% to 100% gives an increase of 119.1%, 105.6% and 96.6% in the water produced for seawater temperatures of 23°C, 28°C and 33°C, respectively.

Figure 5(b) shows the unit production cost obtained by thermo-economic analysis. As seen from the figure, unit cost is decreased when the production load is increased. Also, the increment of seawater temperature provides drop in the unit cost of desalinated water. The minimum calculated production cost of the desalinated water is 1.7 \$/m³ at the

condition of full load and a seawater temperature of 33°C. The high costs related to the rejected brine shows that recovering these expenses may prove to be cost-effective for the plant.

Figure 5 Variation of, (a) distillate production (b) unit production cost as a function of production load for different seawater temperatures (see online version for colours)



4 Conclusions

In this study, energy, exergy and thermo-economic analyses were carried out by considering different operating loads, different seawater temperatures as well as upper brine temperatures. The purpose of the study was to determine the main performance parameters of the related system and also to examine the exergy destroyed in every component of the plant. Besides, the unit cost of distilled water was calculated considering the production load. Main findings of the study are summarised as:

- The enhancement of the production load increased the GOR and exergy efficiency, decrease the specific heat consumption and specific exergy destruction.
- The increasing of production load and seawater temperature are also benefited for the unit cost of distilled water. The unit cost of distilled water was computed to be 1.7 \$/m³ at full load.
- The drop of the upper brine temperatures decreases the specific heat consumption and specific exergy destruction as well as increases the GOR.
- The performance parameters of the desalination plant highly dependent on seawater temperature. The improvement in the seawater temperature positively affected vital performance parameters.
- The steam jet ejector and the effects are the subsystems that primary sources of irreversibilities in the desalination plant. The steam jet ejector controls approximately 45% of the total exergy destruction.
- The most efficient operating point of the system is achieved when the maximum seawater temperature, the maximum production load and minimum upper brine temperature are available.

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References

- Alasfour, F.N. and Alajmi, H.F. (2010) 'Integration of TVC desalination system with cogeneration plant: parametric study', *Fourteenth International Water Technology Conference*, Cairo, Egypt, pp.117–132.
- Alasfour, F.N., Darwish, M.A. and Bin Amer, A.O. (2005) 'Thermal analysis of ME-TVC+MEE desalination systems', *Desalination*, Vol. 174, No. 1, pp.39–61.
- Al-Mutaz, I.S. and Wazeer, I. (2014) 'Comparative performance evaluation of conventional multi-effect evaporation desalination processes', *Applied Thermal Engineering*, Vol. 73, No. 1, pp.1192–1201.
- Askari, I.B., Ameri, M. and Calise, F. (2018) 'Energy, exergy and exergo-economic analysis of different water desalination technologies powered by linear Fresnel solar field', *Desalination*, Vol. 425, pp.37–67.
- Bejan, A., Tsatsaronis, G. and Moran, M. (1996) *Thermal Design and Optimization*, John Wiley and Sons, Canada.
- Beyralvand, D. and Yaghoubi, M. (2018) 'Thermoeconomic analysis and multi-objective optimization of MED-TVC and MED-TVC.FH systems using ABC algorithm', *International Journal of Exergy*, Vol. 26, No. 3, pp.329–358.
- Bin Amer, AO. (2009) 'Development and optimization of ME-TVC desalination system', *Desalination*, Vol. 249, No. 3, pp.1315–1331.
- Brogioli, D., La Mantia, F. and Yip, N.Y. (2018) 'Thermodynamic analysis and energy efficiency of thermal desalination processes', *Desalination*, Vol. 428, pp.29–39.
- Fellah, G., Shuia, E. and Rohoma, A. (2014) 'Exergy assessment for a typical multi-effect thermal vapor compression desalination unit (MED-TVC)', *Journal of Engineering Research*, No. 20, pp.57–66.
- Fitzsimons, L., Corcoran, B., Young, P. and Foley, G. (2015) 'Exergy analysis of water purification and desalination: a study of exergy model approaches', *Desalination*, Vol. 359, pp.212–224.
- Ghaebi, H. and Abbaspour, G. (2018) 'Thermoeconomic analysis of an integrated multi-effect desalination thermal vapor compression (MED-TVC) system with a trigeneration system using triple-pressure HRSG', *Heat and Mass Transfer*, Vol. 54, No. 5, pp.1337–1357.
- Hamed, O.A., Zamamiri, A.M., Aly, S. and Lior, N. (1996) 'Thermal performance and exergy analysis of a thermal vapor compression desalination system', *Energy Conversion and Management*, Vol. 37, No. 4, pp.379–387.
- Khalid, K.A., Antar, M.A., Khalifa, A. and Hamed, O.A. (2018) 'Allocation of thermal vapor compressor in multi-effect desalination systems with different feed configurations', *Desalination*, Vol. 426, pp.164–173.
- Li, Z., Siddiqi, A., Anadon, L.D. and Narayanamurti, V. (2018) 'Towards sustainability in water-energy nexus: ocean energy for seawater desalination', *Renewable and Sustainable Energy Reviews*, Vol. 82, pp.3833–3847.
- Mabrouk, A.A., Nafey, A.S. and Fath, H.E.S. (2007a) 'Thermoeconomic analysis of some existing desalination processes', *Desalination*, Vol. 205, Nos. 1–3, pp.354–373.
- Mabrouk, A.A., Nafey, A.S. and Fath, H.E.S. (2007b) 'Analysis of a new design of a multi-stage flash-mechanical vapor compression desalination process', *Desalination*, Vol. 204, Nos. 1–3, pp.482–500.

- Mohammadi, K. and McGowan, J.G. (2018) 'Thermodynamic analysis of hybrid cycles based on a regenerative steam Rankine cycle for cogeneration and trigeneration', *Energy Conversion and Management*, Vol. 158, pp.460–475.
- Sekar, S. and Saravanan, R. (2011) 'Exergetic performance of ecofriendly absorption heat transformer for seawater desalination', *International Journal of Exergy*, Vol. 8, No. 1, pp.51–67.
- Shen, S., Zhou, S., Yang, Y., Yang, L. and Liu, X. (2011) 'Study of steam parameters on the performance of a TVC-MED desalination plant', *Desalination and Water Treatment*, Vol. 33, Nos. 1–3, pp.300–308.
- Zarzo, D. and Prats, D. (2018) 'Desalination and energy consumption. What can we expect in the near future?', *Desalination*, Vol. 427, pp.1–9.

Nomenclature

B	Brine mass flow rate, (kg/s)
C	Liquid (distillate, feed and brine) specific heat, (kJ/kg.K)
$\dot{C}$	Exergy cost rate, (\$/h)
c	Unit exergy cost, (\$/kJ)
D	Distillate mass flow rate, (kg/s)
E	Exergy rate, (kW)
F	Feed water mass flow rate, (kg/s)
GOR	Gain output ratio
h	Specific enthalpy, (kJ/kg)
I	Irreversibility rate, (kW)
L	Specific latent heat, (kJ/kg)
$\dot{m}$	Mass flow rate, (kg/s)
N	Number of effects
P	Pressure, (bar)
S	Motive steam supply mass flow rate, (kg/s)
s	Entropy, (kJ/kg.K)
T	Temperature, (°C)
<i>Subscripts</i>	
c	Condensate
cw	Seawater
d	Distillate
dc	Steam ejector discharge condition
e	Evaporator
ej	Ejector
ev	Entrained vapour
f	Feed
gn	Saturated vapour generated in the evaporator
in	input
l	Loss
n	Last effect
s	Steam
t	Total
v	Vapour
<i>Greek number</i>	
Υ_{en}	Specific heat consumption, (kWh/m ³)
Υ_{ex}	Specific exergy destruction, (kWh/m ³)
η_{11}	Exergy efficiency, (%)